

SREEJA REDDY YELURU

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PROFESSIONAL SUMMARY

I am a Graduate Research Assistant in AI at DePaul University, where I recently completed my M.S. in Artificial Intelligence with a 4.0 GPA and a Presidential Scholarship. I have 4+ years across machine learning research, software engineering, and data science, and I build and deploy generative AI systems end to end, from retrieval-augmented generation and open-source LLM applications to transformer-based NLP. I pair strong Python and core ML skills with the production discipline I built over three years at Accenture.

EDUCATION

DePaul University, Chicago, IL

June 2026

Master of Science: Artificial Intelligence, 4.0 GPA, Presidential Scholarship

MVJ College of Engineering, Bangalore, India

2021

Bachelor of Engineering: Computer Science, CGPA 8.71

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java, C, C++, JavaScript
- **Generative AI and NLP:** LLMs, open-source models (Llama, Qwen, Hugging Face Transformers), Retrieval-Augmented Generation (RAG), LangChain, Chroma, vector databases, prompt engineering, text generation, transformers
- **Machine Learning and Deep Learning:** PyTorch, TensorFlow, scikit-learn, supervised and unsupervised learning, neural networks, CNN, LSTM, reinforcement learning, computer vision
- **Data:** data cleaning, feature engineering, exploratory data analysis, pandas, NumPy, SQL, Power BI
- **Tools and Platforms:** Git, Docker, Jupyter, Gradio, OpenCV, REST APIs, Hugging Face Spaces

EXPERIENCE

DePaul University

Sep 2025 - Present

Graduate Research Assistant

Chicago, IL

- I implement matrix-based algorithms in Python for **harmonic chain barcodes**, a new topological descriptor for machine learning.
- I build a reproducible research pipeline covering point-cloud ingestion, filtration, boundary and chain matrices, and barcode extraction.
- I benchmark the method against persistent-homology baselines on real datasets, comparing runtime, memory, and classification performance.
- I translate mathematical specifications into clean, tested, documented code and contribute methods text toward a publication.

Tools: Python, NumPy, SciPy, GUDHI, Jupyter, Git

Accenture

Oct 2021 - Aug 2024

Application Development Analyst

Bangalore, India

- I built and maintained a mainframe trading system for Shell Trading across backend, mid-tier, and frontend layers using Java, Python, COBOL, and SQL.
- I optimized SQL queries and caching, improving API response times by **60%**.
- I delivered a self-service troubleshooting module that reduced recurring client support tickets by **20%**.
- I handled deployments and data migrations for production changes, working directly with cross-functional teams, clients, and end users to resolve issues.
- I led code reviews and sprint demos, and kept clear documentation for knowledge transfer.

Tools: IBM Mainframe, COBOL, JCL, Java, Python, SQL

ClearExam

Nov 2020 - Jan 2021

Web Developer Intern

Remote

- I built responsive, cross-device websites for an ed-tech platform, reducing bounce rate by **30%**, and worked with designers to improve UX.

Tools: HTML, CSS, JavaScript

Verzeo

Aug 2020 - Nov 2020

Data Science Intern

Remote

- I built a Random Forest churn model reaching **85% accuracy** and integrated it into a Power BI dashboard.
- I performed exploratory data analysis and feature engineering to surface actionable business insights.

Tools: Python, scikit-learn, pandas, Power BI

PROJECTS

AI Policy RAG

- I built a **retrieval-augmented generation** system that answers AI policy questions over four real government documents and **cites its exact sources**.
- I reworked the generation step to run an **open-source LLM locally**, removing all external API dependency so the demo is reliable and free to run. I deployed it as a live demo.

Tools: Python, LangChain, Chroma, sentence-transformers, Hugging Face Transformers, Gradio

Agentic RAG for CleanTech Question Answering

- I built a **multi-tool retrieval agent** for clean-technology questions, comparing a base RAG pipeline against an advanced agent.
- I gave the advanced agent a Semantic Scholar paper-search tool and a reasoning tool so it can handle multi-step questions.

Tools: Python, LangChain, ChromaDB, GPT-4o-mini, Semantic Scholar API, sentence-transformers

Malicious Prompt Detector

- I trained a stacking ensemble (Logistic Regression, LinearSVC, LightGBM) on **39,000** labeled prompts with TF-IDF features to flag jailbreaks and prompt injection.
- The model reached **95% accuracy and 0.95 F1**, and I deployed it as a live demo.

Tools: Python, scikit-learn, LightGBM, TF-IDF, pandas, NumPy

T5 Text Summarizer

- I **fine-tuned a T5 transformer** for abstractive summarization to generate concise summaries of long documents, and deployed it as a live demo.

Tools: Python, Hugging Face Transformers, T5, PyTorch

Energy Load Forecasting (ComEd)

- I designed a 3-layer CNN-LSTM that forecasts ComEd electricity demand at **1.4% MAPE**, merging **66,000** hourly load rows with weather from 18 stations and using SHAP for feature attribution.

Tools: Python, PyTorch, CNN-LSTM, SHAP, pandas, NumPy

Diabetic Retinopathy Detection

- I built a classical computer vision and ML pipeline (histogram equalization, wavelet and Gabor filters, PCA, SMOTE) with tuned SVM and Random Forest classifiers reaching **85% accuracy**.

Tools: Python, OpenCV, scikit-learn, PyWavelets, SMOTE

PUBLICATION

House Price Prediction Using Machine Learning. International Journal for Research in Applied Science and Engineering Technology (IJRASET), August 2021.